Adithya Gnanasundar

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Education

University of California, Los Angeles — Los Angeles, CA
B.A. in Computer Science and Linguistics — *Expected 2028*
- Coursework: Data Structures & Algorithms, Discrete Mathematics, Multivariable Calculus, Physics for Engineers

Experience

Yekola — Jan. 2025 – Oct. 2025
Software Engineer Intern — *Toronto, ON*
- Built a multilingual chatbot platform supporting Lingala, Amharic, and Swahili, developing backend services, multilingual data-processing workflows, and retrieval-based response systems to support conversations across three low-resource languages
- Engineered reusable pipelines to ingest, clean, transform, and validate multilingual datasets across low-resource languages
- Integrated LLaMA-based models into the chatbot application stack and developed prompting, model evaluation, and user-feedback processing workflows to support iterative releases and ongoing refinement of multilingual chatbot responses
- Implemented a user-feedback system incorporating responses from 200+ native speakers into product release cycles, enabling continuous testing and refinement of multilingual chatbot responses based on feedback from speakers of supported languages

University of Illinois Urbana–Champaign — May 2024 – Dec. 2024
Software Engineer Intern — *Champaign, IL*
- Designed and implemented Proximal Surrogate Generation, a reusable software framework for stress-testing LLM unlearning systems with semantically related surrogate datasets to assess the robustness and reliability of machine unlearning evaluation
- Built scalable data and evaluation pipelines spanning dataset generation, embedding optimization, likelihood filtering, model inference, and automated metric computation to support comparative unlearning experiments across Llama, Qwen, and Zephyr
- Developed distributed experimentation workflows on HPC infrastructure to benchmark NPO and RMU across 3 datasets and 7 metrics, revealing systematic robustness gaps through comparative analysis of machine unlearning evaluations
- Translated experimental findings into reproducible software artifacts for evaluating LLM unlearning systems and contributed to a published research paper examining robustness gaps and improving the reliability of machine unlearning evaluation

University of California, Santa Cruz — Jun. 2024 – Mar. 2025
Machine Learning Intern — *Santa Cruz, CA*
- Developed a neuro-symbolic reasoning framework integrating Prolog-based knowledge graphs with LLM inference pipelines, combining symbolic knowledge and model outputs to improve factual grounding and consistency on multi-step reasoning tasks
- Designed and deployed distributed GPU training and inference workflows using Kubernetes, Docker, and cloud-native infrastructure, enabling parallel experimentation across model configurations for neuro-symbolic reasoning
- Engineered modular evaluation pipelines in PyTorch and Hugging Face Transformers to benchmark multi-step reasoning and factual consistency across open-source LLMs, comparing model performance on structured reasoning tasks

Projects

Semantic Entropy Probes | *Python, PyTorch, Transformers, NLP* — Jan. 2025 – Aug. 2025
- Developed linear interpretability probes to quantify semantic uncertainty across transformer output distributions, examining latent representation alignment in autoregressive language models to analyze how internal representations relate to uncertainty
- Designed evaluation pipelines integrating semantic entropy metrics, sentence embedding models, and generation analysis to benchmark uncertainty and reasoning consistency across model families by comparing generated language outputs

Linguistics Ellipsis Research | *Corpus Linguistics, NLP* — Jun. 2025 – Aug. 2025
- Conducted cross-linguistic analysis of ellipsis phenomena in Turkish, Korean, Mongolian, and Kazakh using large-scale corpus datasets, identifying and comparing syntactic patterns to examine similarities and differences across four diverse languages
- Developed automated extraction, annotation, and preprocessing pipelines to construct structured multilingual corpora following interlinear glossing standards, supporting comparative analysis of ellipsis across four typologically diverse languages
- Presented findings to UCSC faculty researchers; resulting paper accepted for publication in a peer-reviewed linguistics journal

Awards

BlackBoxNLP @ EMNLP 2025 Paper Acceptance — Sep. 2025
- Published research on semantic uncertainty in language models at BlackBoxNLP, EMNLP 2025

3x Hackathon Winner – $1,000 in Prizes — 2025 – 2026
- Beta University Agent-Native Hackathon (1st Place), YC Orange Slice Growth Hackathon, HackWithBay

Technical Skills

Languages: Python, Java, C++, JavaScript, TypeScript, SQL, HTML/CSS
Frameworks: React, Next.js, Node.js, Express.js, Flask, FastAPI, PyTorch
Tools: Git, GitHub, Docker, Kubernetes, Linux, Vercel, Hugging Face
Databases & Libraries: PostgreSQL, MongoDB, Transformers, NumPy, pandas, scikit-learn